

EM 670: *Design: Manufacture & Assembly*

Course Syllabus

Course Description This course investigates the management of product and process design activities. It is intended for anyone involved in product design, manufacturing engineering, process engineering, or manufacturing management. The course investigates design for manufacturing and assembly, quality function deployment, value analysis, failure modes and effects analysis, error/mistakeproofing, team dynamics and design for 'X'.

Course Objectives Upon completion of this course, students will be able to:

- Identify and apply DFMA tools and principles.
 - Analyze a product's design to evaluate its suitability for assembly and identify improvements to make assembly easier by applying DFA principles
 - Identify the use of DFM/DFA principles in existing products
 - Identify possible modifications to existing products by applying DFA /DFM principles
 - Complete team based DFM/A project per prior objectives.
 - Design basic error proofing methods for assembly operations.
 - Apply basic DFX (Reliability, Finish, Heat Treatment etc) principles to improve design
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Schedule **Tentative:**

Unit 1 DFM

From Text Book: Chapter 1 - 3

Articles: "Toyota's Set Based Concurrent Engineering" and

"Another Look at how Toyota Integrates Product Design"

These articles are in doc sharing.

Unit 2 DFM

From Text Book: Chapter 4-7

Unit 3 DFA

From Text Book: Chapter 21 - 25

Unit 4: Tools for DFMA

From Text Book: Chapter 32 A - G

Unit 5 Design for Finish, Heat Treatment, Marking, Low Quality

Production

From Text Book: Chapter 26 – 28 and 31

Unit 6: DFX, Design for Electronics, (Reliability and Error Proofing) Design for Sustainability,

From Text Book: Chapter 29 -30; "Design for the Environment" article in doc sharing

Unit 7: Life Cycle Analysis and Team Dynamics

5-SPD-Lifecycleexample.pdf

"Organizing for DFM" article

Unit 8 : Projects

Final Exam

Project

Communication The principle form of communication in this class is email. Please check your "emich" email account daily. If you prefer a different email address please set your emich account to automatically forward email to your preferred email address.

E-mail me at: mahmed6@emich.edu

Also, In the subject line of your email messages, include the course number, initials of your name, and the subject of the message. **This is very important.** Many instructors teach multiple courses and quick visual differentiation is important.

Grading Policy Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A

92.99 to 90% = A-

89.99 to 87% = B+

86.99 to 83% = B

82.99 to 80% = B-

79.99 to 77% = C+

76.99 to 73% = C

72.99 to 70% = C-

(no D grades, based on Grad School requirements)

69.99 and below = E

Weightage

You will be graded on Assignments, Exams, Discussions, Debate, Research paper and presentation according to the following weightage:

Individual Assignment = 20%

Team Assignment = 20%

Discussions = 10%

Exams = 10%

DFMA Final Project = 25%

DFMA Mini Project = 15%

Final course grades will be based primarily (*but not necessarily only*) on the quantitative scale shown above--but are subject to adjustment based on the judgment of your professor in consideration of qualitative factors, including academic dishonesty and displays of other unethical, unprofessional, or uncivil behaviors that are unbecoming of a good engineering manager and graduate student. (*You must to be prepared for the reality of professional life in business and industry--there is usually more to getting a good performance evaluation from your boss and peers than just production of work, other factors include: (a) personal professional skills, (b) professional communication skills, and (c) ethics and emotional intelligence.*) (a) (b) and (c) above are of vital importance to becoming successful as an Engineering Manager, and why they are three of our EM Program Student Learning Objectives. We expect to see them demonstrated by students in our classes.

General Policies

A. Problems.

Students should discuss problems or mis-understandings about their course work directly with the instructor via telephone or email, not via broadcast emails to the entire class.

B. Online Participation and Conduct.

Since there are no required face-to-face meetings for online courses, interaction via threaded discussions is often very important for effective learning. As such, good class participation is essential and will account for a significant part of your grade. Plan to contribute to every discussion topic with a "quality" contribution -- i.e., a response that expresses clear thinking and that is relevant to the discussion. This usually involves more than just "I agree."

Our instructors and the University can not and will not be responsible for the content of any personal messages that are sent from one student to other students via email or threaded discussion. To maintain privacy, it is best NOT to share phone numbers or addresses to class mates unless necessary for

team communications. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using email, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.

When you send emails related to a class, you should put informational codes in the subject line of your email messages. For example, include the: **initials of your full name, course number**, and the **subject of the message** (such as the name of an assignment that you may be asking about); such as, Subject: **XYZ670Homework1**. Also, do not forget to include your full name in the text of your email message.

C. Grammar, Spelling, Writing Style, and Format.

Good grammar, spelling, writing style, and APA format is required for formal papers, per School of Engineering Technology (SET) policy. Students should follow APA (American Psychological Association) technical writing style and format guidelines for papers submitted in all SET classes. Some professors will emphasize use of APA guidelines more than others. It is recommended that you get a copy of the 5th edition of the APA Publication Manual as soon as possible, as you will use it throughout your degree. Unless otherwise specified by your professor, you should write your papers in final manuscript format (see Chapter 6, APA manual), NOT copy manuscript format (which is used when submitting articles to journal editors for pre-publication review and editing. You should especially be familiar with traditional and electronic referencing guidelines for quoting, paraphrasing, and referencing information that you use, to help avoid Academic Dishonesty problems, as mentioned above.

See <http://www.apa.org/journals/acorner.html#pubmanual> for more specific information and examples.

You should consider all of your communications with the class and professor as a professional communication. Good grammar and spelling is important for a professional in the business world. If that is difficult for you, we suggest that you compose threaded discussion responses or email messages in a word processing program where you can check spelling and grammar, and revise your writing. Then you can cut and paste the work into the threaded discussion or email message.

D. Academic Dishonesty.

The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "E" for the course in which the activity occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work, papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or taking credit for another's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

Our managerial approach regarding minor occurrences of academic dishonesty in this course is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an example of the problem with the rest of the class, as a learning experience for the whole class. The **consequences may be more severe** (such as failure of the assignment or the entire course), and without a warning, for major occurrences and/or second minor occurrences of academic dishonesty that occur in this course, or that are on record for other courses that you may have taken. Case records of academic dishonesty are kept in the SET department and/pr the Student Judicial Services office.

I will be using TurnItIn.com to check papers for originality.

E. Incomplete Final Grades (I): will be given only in accordance with EMU's guidelines, which are stated in the 2005-2007 Graduate Catalog (p.19), at <http://www.emich.edu/public/catalogs/2005-2007/grad/20051gfront.pdf>, as follows: I (Incomplete) Grades - An I grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course (50% or more) merits a grade of B or better.

F. Student and Exchange Visitor Information Service (SEVIS).

For students who hold an F or J visa, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), 244 Student Center, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government -- Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services -- SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003). OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

Per SEVIS regulations, International students who hold an F or J visa can only take one fully online class (no face-to-face meetings) a semester. Hybrid classes have face-to-face meetings, so are allowed.

G. Observance of Religious Holidays.

Eastern Michigan University recognizes the right of students to observe religious holidays without penalty to the student. We have students from a wide variety of cultures and countries in our EM program. We will follow a

functional policy for this class that can accommodate everyone's desire to observe a religious holiday. Students in EM 670 do not have to do class work on their religious holiday, but deadline dates for assignments will not be extended. If a student's religious holiday occurs on the last day of a unit, the student should not wait until the last day of the unit to complete their assignments. Per course policy, late assignments will not be accepted. Students are encouraged to not wait until the last minute to submit assignments.

H. Personal Involvement and Time.

Students often have the misperception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier!

Be aware of the general guidelines to professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so we expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10-15 (or more) hours of your time each week.

This is an Engineering Management course, part of the EM program, taken by EM masters and doctoral students. Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Poor planning and procrastination, such as trying to complete all work at the last minute, leads to poor quality and missed deadlines because there is no buffer available to deal with unexpected problems. Many of our EM students lead very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program can not organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. As such, we will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.

Course Policies and Procedures

A. Access to Course.

Our course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two full weeks with the exception of Unit 1, which starts on a Wednesday, and Unit 8, which ends on a Friday. **All assignments will be due at the end of each unit on Sunday nights, by 11:59 PM ET, except for selection of presentation topics in Unit 1 (due earlier), and the submission of formal papers, which must**

be submitted by the Sunday night that is one week prior to the opening of unit of that presentation to the class. No assignments required to be completed for a unit will be accepted after the close of that unit. All completed units will remain open for review, until the end of the semester.

B. Email to Instructor.

For EM 695 course-related submissions and communications, please send all email directly to Dr. Sohail at his personal email account, at mahmed6@emich.edu. Also, in the subject line of your email messages, include the: **initials of your full name**, **course number**, and the **subject of the message** (such as the name of an assignment that is attached to your email); for example, Subject: **XYZ69Design1**.

I can not be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 695-related email before 12 noon on several days during the week. Due the amount of email I will receive from students, as well as other work and travel commitments, I may not be able to respond to emails or confirm receipt of submitted assignments more than once or twice a week--although I will try to do so more frequently. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor a fellow student may be able to respond quickly enough to help you. Also, it make take a day or two before you receive an emailed confirmation from me that I received your submitted assignment. Typically, the confirmation will read: "Got it." or "Got it, Dr. Sohail."

C. Course Format and Activities.

The general format of this class will be through online guidance from the professor and students' submission of assignments, which include: reading presentation assignments and quizzes, case study discussions, peer evaluations of fellow students' work, engineering ethics project work, a leadership personality profile assessment and report and taking a final exam.

D. Assignment Software, Submission to Drop Box, and File Names.

There are so many different word processing programs that reading assignments can become a problem in online courses if there is no standard policy set for them. **It is preferred that you submit papers in MS Word file format.** Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message--although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2007 format. You may also send single or multiple files in .zip compressed format.

Submit assignments to the Unit Drop Box as directed for each Unit. The file names used for assignments should follow a coding format similar to that used for course-related emails. For example, include the: initials of your full name, course number, and the name of the assignment with its file type; therefore, the file name of an

assignment would look like: **XYZ695Presentation1.doc.**

I will use TurnItIn.com to check papers for originality after they are submitted to the drop box.

E. Assignment Delivery and Technical Problems.

Since you will have more than two weeks and three weekends available for completing your work throughout most of the course, we expect you to get it done on time. **No assignments will be accepted after the close of a unit.** Written paper assignments are to be submitted using the Unit Assignment Drop Box, and are due one week prior to the close of a unit (see schedule in syllabus). Written paper assignments will not be accepted after the close of a unit. Other work such as threaded discussions, quizzes, and exams are submitted using those course tools, which automatically close at the end of a unit. The final exam (and any other assignments due in our Unit 8 final week) will not be accepted late. We will provide early access to units for you. Plan on getting your work done during the first half of the unit, and plan on encountering unexpected problems. **You have been made aware of the consequences of poor planning, procrastination, and not getting your work done on time.**

In some cases your computer may freeze or you might be having issues with the eCourse or eCompanion web site (your exam or quiz or assignment not saved or your being kicked out from the page etc), in all such cases contact eCourse/eCompanion help desk and copy your instructor on all such emails. If help desk confirms that the issues was related to them then you instructor will give you extra time to finish all such assignments. So please make sure that you don't wait until last day or last hour to finish up your assignments.

We encourage you to make hard copies of your assignments and communications with your instructors. Computer crashes and disasters can happen--we expect you to practice good risk management behavior, just like good engineering managers. We do not view ISP problems as an excuse for turning work in late. Alternative methods of submitting assignments include:

1. Fax it to Dr. Sohail at 734.487.8755 (EMU).
2. Overnight express hard copy and on disk.
3. Make arrangements over the phone with me.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public Library. Our guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate.

International Students

Attention students who hold an F or J visa: To avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), King Hall, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective

January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003). OIS enters the data into a real-time database and new documents or I-20 s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.